



# Construction of a translational whole-body physiologically based pharmacokinetic model to investigate the biodistribution of lipid nanoparticle-based mRNA therapeutics for intracellular protein targets

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## Abstract

Lipid nanoparticle (LNP)–based messenger RNA (mRNA) therapeutics enable in vivo production of diverse protein modalities, but their translation is complicated by multiscale pharmacokinetics encompassing tissue transport, cellular uptake, and intracellular processing. We developed a platform translational whole-body physiologically based pharmacokinetic (PBPK) model to mechanistically characterize the biodistribution and protein expression of intravenously administered mRNA–LNP therapeutics. The model integrates receptor-mediated, saturable tissue infiltration, lymphatic recirculation, hepatic uptake, and explicit intracellular processes including endosomal degradation, mRNA escape, translation, and protein turnover. Parameters were calibrated using comprehensive rat biodistribution data containing both tissue-level mRNA and translated protein measurements, enabling characterization of rapid early tissue deposition and dominant hepatic protein production. Sensitivity analyses identified endosomal degradation, mRNA stability, and translation efficiency as the primary determinants of hepatic protein exposure, whereas increasing tissue influx alone yielded minimal effect. Using fundamental allometric scaling principle, the model was translated from rat to human and evaluated against published clinical plasma mRNA pharmacokinetics of an mRNA-encoded monoclonal antibody. Virtual population simulations reproduced dose-dependent peak timing and clearance across multiple dosing regimens. Cell-type specific simulation analyses further demonstrated the competing roles of Kupffer cells and hepatocytes in shaping liver protein exposure. This whole-body PBPK framework provides a mechanistic and predictive platform to support first-in-human dose selection, to interrogate multilevel determinants of mRNA pharmacokinetics, and to guide rational mRNA–LNP design.

**Keywords** Physiologically based Pharmacokinetic (PBPK) modeling · Lipid nanoparticle (LNP) mRNA therapeutics · Translational pharmacokinetics · Model-informed drug development (MIDD)

## Introduction

Messenger RNA (mRNA) technology has emerged as a transformative and increasingly established modality in modern therapeutics and vaccine development, offering a versatile platform for targeted protein expression delivered via lipid nanoparticles. This is exemplified by the unprecedented speed and scale with which mRNA based vaccines were developed against SARS-COV-2 during the COVID-19 pandemic [1, 2]. Similar progress has been achieved with neoantigen mRNA cancer immunotherapy [3, 4], highlighting the versatility of the platform for both infectious disease and oncology vaccine applications. Beyond applications to vaccine development, the mRNA technology is

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being explored for therapeutic purposes, such as development of mRNA encoded monoclonal antibodies [5], bispecific antibodies for oncology [6], in situ CAR-T (chimeric antigen receptor T cell) therapy for cancer [7], and mRNA based enzyme replacement therapy for rare diseases [8, 9]. Achieving success in the broader application of lipid-based nanoparticle mRNA (mRNA-LNP) therapeutics requires a quantitative understanding of how LNP based mRNA constructs distribute across tissues, traverse cellular barriers, escape the endosomal pathway and translate into functional protein over clinically relevant dose regimens and time scales.

The pharmacokinetics of mRNA-LNP is complex and multi-layered. After intravenous (IV) administration, the particles encounter a dynamic protein corona that determines opsonization, clearance by the mononuclear phagocyte system and interaction with apolipoprotein E (ApoE) dependent receptors on hepatocytes and endothelial cells [10, 11]. Particle size, ionizable-lipid pKa and helper-lipid composition all modulate this corona and hence tissue tropism [12, 13]. Within target organs the particles must cross the endothelial fenestrae, undergo uptake (often via scavenger or LDL-family receptors), survive endosomal degradation and finally release mRNA into the cytosol where ribosomes translate the mRNA into encoded protein. Each layer involves distinct rate-limiting steps and can introduce pronounced species differences [14]. Consequently, while conventional allometric approaches remain valuable tools, relying on them without accounting for the absorption, distribution, metabolism, and excretion (ADME) complexities of mRNA-LNP may lead to less informative or potentially misleading dose projections.

Physiologically based pharmacokinetic (PBPK) models provide a mechanistic framework to integrate those layer-specific processes with species-specific anatomy as well as blood and lymphatic flow [15]. Platform PBPK approaches are established for small molecules and large molecule therapeutics (e.g. monoclonal antibodies, ADCs, etc.), where they have enabled various decision makings, such as first-in-human (FIH) dose selection, drug-drug interaction assessment and regulatory filings [16, 17]. Emerging in parallel, PBPK frameworks for mRNA-LNP are nascent. Existing efforts span models focused on the liver [18], organ-specific perfusion models [19], integrated minimal PBPK/QSP models within the context of rare diseases [20], and whole-body PBPK models used to simulate exposure of secreted biologics such as bispecific T cell engagers [21]. Our goal was to develop a whole-body PBPK modeling platform to simultaneously capture (i) receptor-mediated, saturable trans-endothelial transport, (ii) intracellular trafficking and translation, and (iii) rigorous inter-species scaling across all major organs.

Recent biodistribution studies highlight why these features are critical. Ci et al., [22] showed that plasma and liver mRNA concentrations peak within 10–15 min of dosing, a profile incompatible with perfusion-limited tissue uptake. Sago et al., [23] demonstrated that sub-cellular competition between Kupffer cells, liver endothelial cells, and hepatocytes shape functional delivery of lipid nanoparticles into the liver tissue. Meanwhile, recent data reveal steep pH-dependent changes in ionizable lipid structure that alter endosomal escape kinetics [24]. Collectively, these findings argue for development of a translational PBPK model that can disentangle early, receptor-driven deposition from downstream intracellular bottlenecks and present an opportunity to predict human mRNA-LNP PK from preclinical species.

Here we report the construction and validation of a whole-body translational PBPK platform for mRNA-LNP therapeutics. In this modeling framework, pharmacokinetics (PK) refers specifically to mRNA disposition, whereas pharmacodynamics (PD) refers to the resulting encoded protein expression. Building on the anatomical template of Shah and Betts [15], we introduced Michaelis–Menten infiltration terms to model receptor saturation and receptor mediated tissue infiltration, an endosomal sub-compartment to account for degradation and escape, and mRNA to protein translation within each tissue. Parameters were calibrated against the comprehensive rat dataset of Ci et al., [22], encompassing both mRNA and translated protein across different organs. Although the calibration was performed for a specific LNP formulation, the underlying model structure is designed to support re-calibration for other LNP designs, recognizing that differences in chemical composition may alter tissue distribution, cellular uptake, and downstream expression. The current implementation focuses on intravenous (IV) administration, consistent with the available biodistribution data. The model was then allometrically scaled to human based on fundamental allometric scaling principles. This allometric scaling was shown to enable adequate prediction of human plasma mRNA PK. Virtual patient simulations reproduced published plasma PK profiles for an mRNA encoded monoclonal antibody targeting Chikungunya virus at doses from 0.1 to 0.6 mg/kg [5], providing confidence in cross-species scaling.

Beyond recapitulating observed pharmacokinetics, the model enables hypothesis-driven exploration of LNP and mRNA dependent features. Local and global sensitivity analyses pinpointed endosomal degradation, mRNA stability and receptor-mediated influx as the dominant determinants of hepatic protein exposure—a finding consistent with empirical structure–activity relationships linking ionizable lipid pKa and in vivo potency [12].

Simulations further clarified how modulating Kupffer-cell versus hepatocyte uptake shifts the balance between clearance and functional protein expression, informing strategies to retarget LNPs to different cell types.

In sum, our expanded PBPK model closes a critical gap in the predictive toolkit for mRNA medicines. By coherently representing whole-body disposition, intracellular processing and inter-species scaling, it offers a rational platform for quantitatively understanding the ADME of mRNA-LNP across species, enabling first in human (FIH) dose selection, candidate prioritization and the engineering of next-generation LNPs with improved tissue selectivity.

## Methods

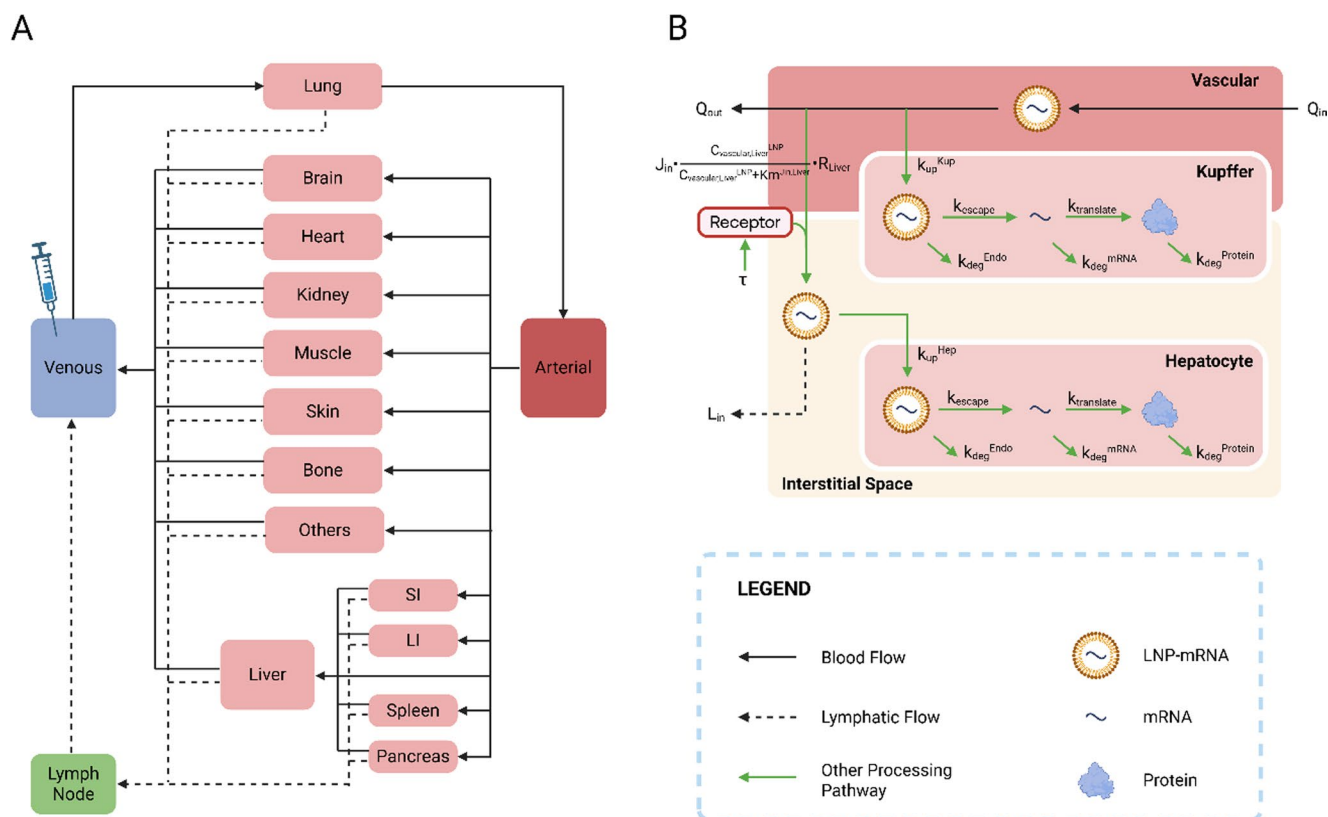
### Model structure

The schematics of the whole-body PBPK model is shown in Fig. 1A. The model consists of venous and arterial compartments and thirteen organ compartments (lung, brain, heart, kidney, muscle, skin, bone, small intestine, large intestine, spleen, pancreas, lymph nodes, and liver). Following IV

injection, mRNA-LNP enters the venous compartment, from which it travels to the rest of the tissues through arterial flow. From each organ, mRNA-LNP can return to the venous pool via blood flow or enter the lymphatic system through the lymph node, ultimately re-circulating back to the venous compartment. Within portal organs (small intestine, large intestine, spleen, and pancreas), mRNA-LNP flows into the liver before re-entering the venous circulation. Blood flow rates and organ volumes are set to species specific physiological values [15], with the details provided in the [supplementary material](#).

### Sub-compartment model structure

Each organ compartment is further divided into three sub-compartments: vascular space, interstitial space, and cellular space. mRNA-LNP enters each tissue compartment through vascular blood flow ( $Q_{tissue}$ ) and returns to the venous compartment at a rate determined by the difference between blood flow and lymphatic flow ( $Q_{tissue} - L_{tissue}$ ). Within the vasculature, mRNA-LNP transports into the interstitial space via the influx rate ( $J_{in}^{tissue}$ ). To account for receptor-mediated rapid uptake of mRNA-LNP observed in biodistribution data, we incorporated a saturating function



**Fig. 1** Schematic of the whole-body PBPK model (**A**) and sub-compartments in liver tissue (**B**). Blood flows are indicated by black solid arrows, lymphatic flows are depicted by dashed black arrows, and

other cellular processes were presented by green solid arrows. SI: small intestine. LI: large intestine

into the tissue infiltration process. The apparent  $K_m$  of the infiltration function ( $K_m^{Jin}$ ) was assumed to be uniform across the tissues, estimated using plasma mRNA PK data. Additionally, a probability function was included to represent receptor depletion and regeneration. Receptor depletion is governed by the depletion rate ( $k_{receptor}$ ), while receptor recovery half-life is assumed to be 72 h ( $\tau$ ), long enough to capture the observed mRNA PK profile. Example equations for receptor-mediated tissue infiltration are provided in the Model Equations section, with the full set of equations and reactions available in the supplementary excel file associated with this manuscript.

mRNA-LNP within the interstitial space can either be taken up by cells with an uptake rate ( $k_{up}^{tissue}$ ) or flow into the lymph node through lymphatic flow. The lymphatic flow is assumed to be 0.2% of the plasma flow rate ( $L_{tissue} = Q_{tissue}/500$ ) and is limited by the lymphatic reflection coefficient ( $\sigma$ ), consistent with the assumptions used in published study [15]. Within the cellular compartment of each tissue, mRNA-LNP undergoes endosomal degradation at a rate  $k_{deg}^{LNP}$ , while mRNA escapes from LNP and endosome at the cumulative escape rate of  $k_{escape}$ . Escaped mRNA within the cytoplasm either undergoes degradation at the degradation rate of  $k_{deg}^{mRNA}$  or is translated to produce the desired protein at the translation rate of  $k_{translate}$ . The cumulative escape rate of mRNA from LNP and endosome is assumed to be the same across different tissues and is fixed to a value reported in the literature [18]. The tissue influx rates, cellular uptake rates, endosomal degradation rates, mRNA degradation rates, and protein translation rates were estimated by calibrating the model to the Rat biodistribution data [22]. Unlike other tissue compartments, the liver compartment includes two distinct cell types: Kupffer cells, which are adjacent to the vasculature and directly take up mRNA-LNP from the vasculature, and hepatocytes, that reside within the interstitial space. The structure of the liver sub-compartments is shown in Fig. 1B, the structure of the non-liver sub-compartments can be found in Supplementary Materials Fig. 1, and the list of parameters is provided in the [supplementary material](#).

### Rat to human scaling of the PBPK model

The calibrated model was scaled to human to explore human PK projection. The results were validated by the published clinical mRNA PK data [5]. Physiological parameters, including organ tissue volumes and blood flow rates, were adjusted based on values for a 71 kg adult human [15]. The

tissue influx rates were scaled according to the allometric scaling law, using a scaling exponent of 0.75:

$$J_{in, Human}^{Tissue} = J_{in, Rat}^{Tissue} \left( \frac{71000}{280} \right)^{0.75}$$

The apparent  $K_m$  for tissue influx was assumed to be the same as in rat. The cellular uptake rates were allometrically scaled with a scaling exponent of  $-0.25$ :

$$k_{up, Human} = k_{up, Rat} \left( \frac{71000}{280} \right)^{-0.25}$$

The mRNA escape rate, endosomal degradation rate, and mRNA degradation rate were assumed to be similar across species. The scaled human model was then used to perform virtual trial simulations and the results were compared with clinical plasma mRNA PK data from the mRNA encoded mAb targeting Chikungunya virus [5].

### Model equations

In this section, we present examples of the model equations used in this study. The model was implemented in MATLAB SimBiology (MathWorks Inc., 2022). All the analyses, including sensitivity analysis and parameter fitting, were performed in MATLAB. Global sensitivity analysis was performed using the partial rank correlation coefficient (PRCC) method as described in [25]. The full set of model equations, reactions, and parameters is provided in the supplementary excel file associated with this manuscript.

$$\begin{aligned} \frac{dM_{Venous}^{LNP}}{dt} = & \underbrace{-k_{up} V_{Venous}^{mac} \cdot C_{Venous}^{LNP}}_{\text{Uptake by venous macrophage}} - \underbrace{Q_{Lung} \cdot C_{Venous}^{LNP}}_{\text{Transport to lung}} \\ & + \underbrace{(Q_{Liver} - L_{Liver} + Q_{PO} - L_{PO}) C_{Liver,vascular}^{LNP}}_{\text{Transport from liver}} \\ & + \underbrace{L_{LN} \cdot C_{LN}^{LNP}}_{\text{Transport from lymph node}} + \text{other terms} \end{aligned} \quad (1)$$

Equation 1 describes the rate of change in the total amount of mRNA encapsulated in LNP in plasma. The first term represents cellular uptake, which is proportional to the volume of macrophages in plasma. The mathematical form of the uptake equation follows a similar structure as in [20, 26]. The second term accounts for flow out of the venous compartment ( $Q_{Lung}$ ). The last term represents flow from the lymph node into the venous compartment

with the lymphatic flow rate ( $L_{LN}$ ), other terms denote plasma flow rates from tissues other than liver (heart, skin, muscle, etc.).

$$\begin{aligned} \frac{dM_{Liver,vascular}^{LNP}}{dt} = & \underbrace{Q_{Liver} C_{Arterial}^{LNP}}_{\text{Transport from arterial}} - \underbrace{(Q_{Liver} - L_{Liver}) C_{Liver,vascular}^{LNP}}_{\text{Transport to venous}} \\ & + \underbrace{(Q_{SI} - L_{SI}) C_{SI,vascular}^{LNP}}_{\text{Transport from small intestine}} + \underbrace{(Q_{LI} - L_{LI}) C_{LI,vascular}^{LNP}}_{\text{Transport from large intestine}} \\ & + \underbrace{(Q_{spleen} - L_{spleen}) C_{spleen,vascular}^{LNP}}_{\text{Transport from spleen}} \\ & + \underbrace{(Q_{pancreas} - L_{pancreas}) C_{pancreas,vascular}^{LNP}}_{\text{Transport from pancreas}} \\ & - \underbrace{J_{in}^{Liver} \frac{C_{Liver,vascular}^{LNP}}{C_{Liver,vascular}^{LNP} + K_m^{Jin,liver}} \cdot R_{Liver} \cdot C_{Liver,vascular}^{LNP}}_{\text{Receptor mediated tissue infiltration}} \\ & - \underbrace{k_{up}^{Liver} \cdot C_{Liver,vascular}^{LNP} \cdot Vol_{cell}^{Kupffer}}_{\text{Cellular uptake of Kupffer cell from plasma}} \end{aligned} \quad (2)$$

Equation 2 describes the rate of change in the total amount of mRNA encapsulated in LNP in the liver vasculature space. The second-from-last line represents tissue infiltration, which depends on the saturation function and the probability of receptor availability ( $R_{Liver}$ ). The remaining terms describe flow from the portal organs (small intestine, large intestine, spleen, and pancreas) and flow into the vascular compartment.

$$\frac{dR_{Liver}}{dt} = \underbrace{-k_{receptor} \cdot \frac{C_{Liver,vascular}^{LNP}}{C_{Liver,vascular}^{LNP} + K_m^{Jin,liver}} \cdot R_{Liver}}_{\text{Receptor depletion by tissue infiltration}} + \underbrace{\frac{1 - R_{Liver}}{\tau}}_{\text{Receptor recovery}} \quad (3)$$

Equation 3 describes the rate of change in probability of receptor availability in the liver vascular space for tissue infiltration. The baseline value of the probability of receptor availability is set to 1. The first term represents receptor depletion as LNP infiltrates into the tissue, while the second term accounts for receptor recovery, with a half-life set to 72 h.

## Virtual population simulations

A virtual population was created to project human PK through a virtual population simulation with human-scaled model. Key parameters, including tissue influx rates, apparent  $K_m$  of tissue penetration  $K_m^{Jin}$ , cellular uptake rates, endosomal/mRNA degradation rates, escape rates, and translation rates, were varied based on a log-normal distribution with 30% variability. The virtual population demographics, such as body weight,

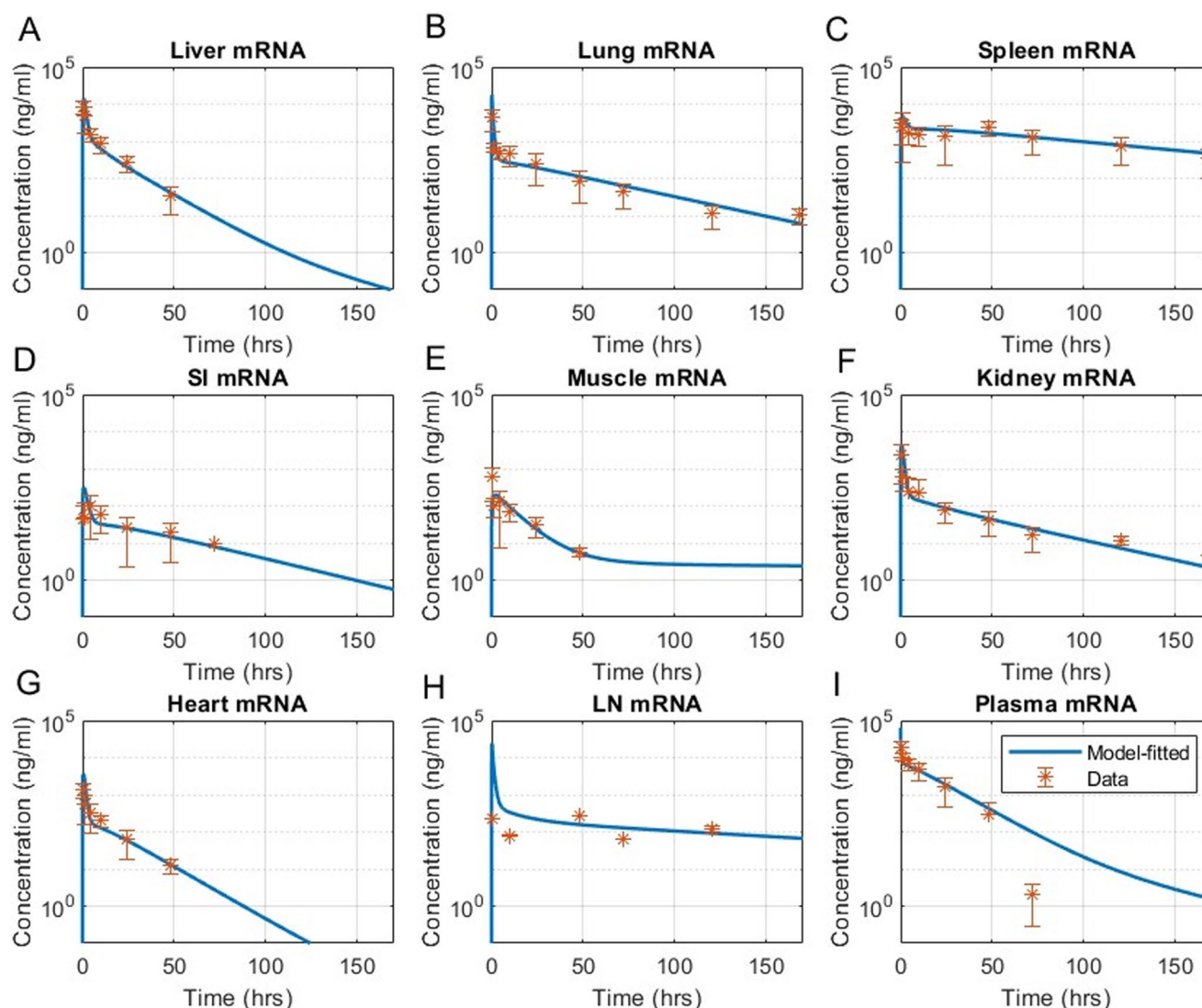
tissue weights, were kept the same. A total of 1000 virtual patients were generated for the simulation.

## Results

### Model calibration to the rat biodistribution data

The rat-scaled model, established based on physiological parameters for a 280-g rat, was calibrated to rat biodistribution data in [22]. The study includes biodistribution data for an intracellular mRNA therapy, in which the translated protein remains within the cell. Quantified mRNA and protein concentrations across different tissues were measured using a branched DNA (bDNA) assay and an LC-MS/MS assay, respectively, following a 2 mg/kg dose of mRNA administered via 10-minute infusion. Model calibration steps were as follows: we first calibrated the liver mRNA concentration profile to estimate the tissue influx rate, mRNA degradation rate, and LNP endosomal degradation rate. Spleen and lung mRNA data were then calibrated by fitting their respective influx rates as well as mRNA and LNP degradation rates. Plasma mRNA data was then calibrated by fitting  $K_m^{Jin}$  for tissue influx (the apparent  $K_m$  of the receptor availability function), receptor depletion rate ( $k_{receptor}$ ), and cellular uptake rate. Both  $K_m^{Jin}$  and  $k_{receptor}$  were assumed to be the same across all organ types, while cellular uptake was assumed to be the same for all non-liver organs. The model was then calibrated to mRNA data for the remaining tissues, including the small intestine, muscle, heart, and lymph node, by fitting their corresponding tissue influx rates, endosomal degradation rates, and mRNA degradation rates. The summary of calibration results is provided in Fig. 2. Since no protein was detected in the lymph node, the escape rate was set to zero, and only endosomal degradation was estimated. For tissues lacking data (large intestine, brain, skin and bone), the tissue influx rate was assumed to be the same as the lowest estimated influx rate, corresponding to the small intestine. The model calibration results for mRNA biodistribution data are shown in Fig. 2. While the specific product names were not disclosed, the preclinical and clinical data used for model calibration and validation employed similar LNP formulations, with consistent lipid compositions and comparable lipid-to-mRNA ratios, thereby supporting the translational applicability of the model.

The model was then calibrated to the protein data in each tissue location to estimate the translation and intrinsic protein degradation rate. The summary of protein calibration results is provided in Fig. 3. The list of parameters is given in Table S1.



**Fig. 2** Overlay of model calibration (blue solid lines) to mRNA biodistribution raw data at different tissue sites (shown as asterisks)

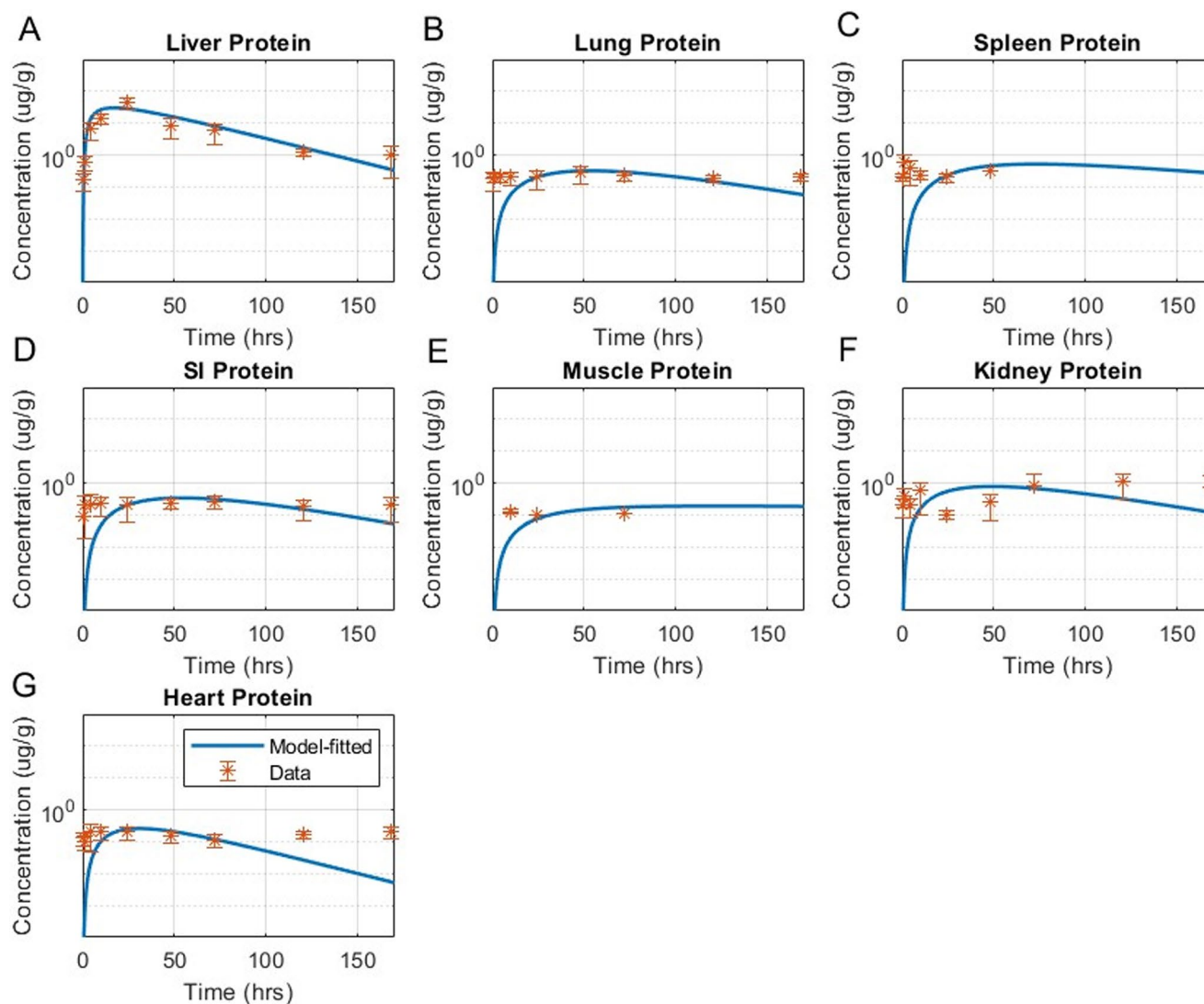
### Data driven analysis of mRNA and protein distribution

Biodistribution raw data was first analyzed to confirm the primary sites of mRNA deposition and protein production. The amount of mRNA in each tissue, calculated from the tissue concentration and corresponding tissue volume, at  $t=10$  min was plotted as percentage of the injected dose (Figs. 4A). Initial biodistribution analysis revealed that at  $t=10$  min (the earliest time-point assessed) the majority of mRNA was in plasma ( $\sim 30\%$  of the injected dose) and the liver ( $\sim 24\%$  of the injected dose). Interestingly, a substantial amount of mRNA was also detected in the muscle tissue ( $\sim 14\%$ ), whereas little mRNA was found in the rest of the tissues in the study ( $<2\%$ ). The rest of the mRNA ( $\sim 30\%$ ) at  $t=10$  min was most likely degraded. This biodistribution analysis was also compared with model simulations,

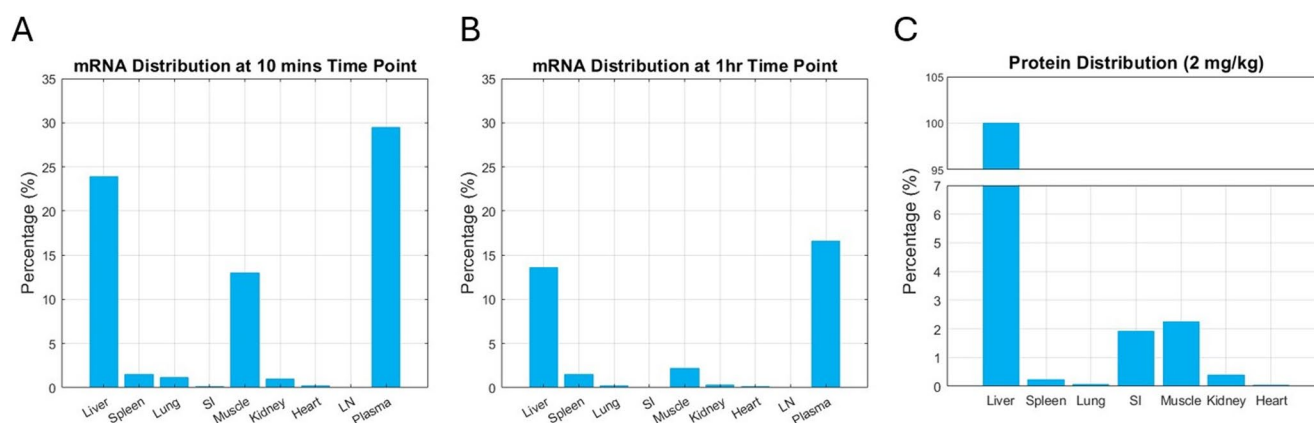
which demonstrated consistent biodistribution behavior (the results shown in Supplementary Figure S2).

Biodistribution of mRNA, given as percent fraction of the injected dose, was calculated at  $t=1$  h across different tissues (Figs. 4B). Most of the remaining mRNA was in plasma and the liver (16% and 14% respectively). However, distribution to the muscle was significantly decreased compared to the level at  $t=10$  min ( $\sim 2.5\%$ ), while the Spleen retained  $\sim 1\%$  of the injected mRNA. These tissues accounted for  $\sim 34\%$  of the injected mRNA indicating significant mRNA clearance within 1 h of IV injection.

Maximum protein production in different tissues relative to the liver protein  $C_{max}$ , was plotted to identify the primary site(s) of protein production (Fig. 4C). Based on the summed protein production across all tissues, the data demonstrate that the liver is the primary site of the protein production accounting for  $\sim 95\%$  of the total protein. A



**Fig. 3** Overlay of model calibration (blue solid line) to the protein biodistribution raw data at different tissue sites (shown as red asterisks)



**Fig. 4** Bar plot of mRNA and protein biodistribution Following IV Administration of 2 mg/kg mRNA-LNP in Rats. (A) Biodistribution of mRNA at  $t = 10$  min at different tissue locations computed as percent injected dose. (B) Biodistribution of mRNA at 1 h time point at different

tissues given as the percent of injected dose. (C) Biodistribution of protein, expressed as tissue-specific protein  $C_{max}$  relative to liver protein  $C_{max}$

similar analysis was performed using protein Cavg, which also indicates the liver as the primary site of protein production (results provided in the Supplementary Material Figure S3).

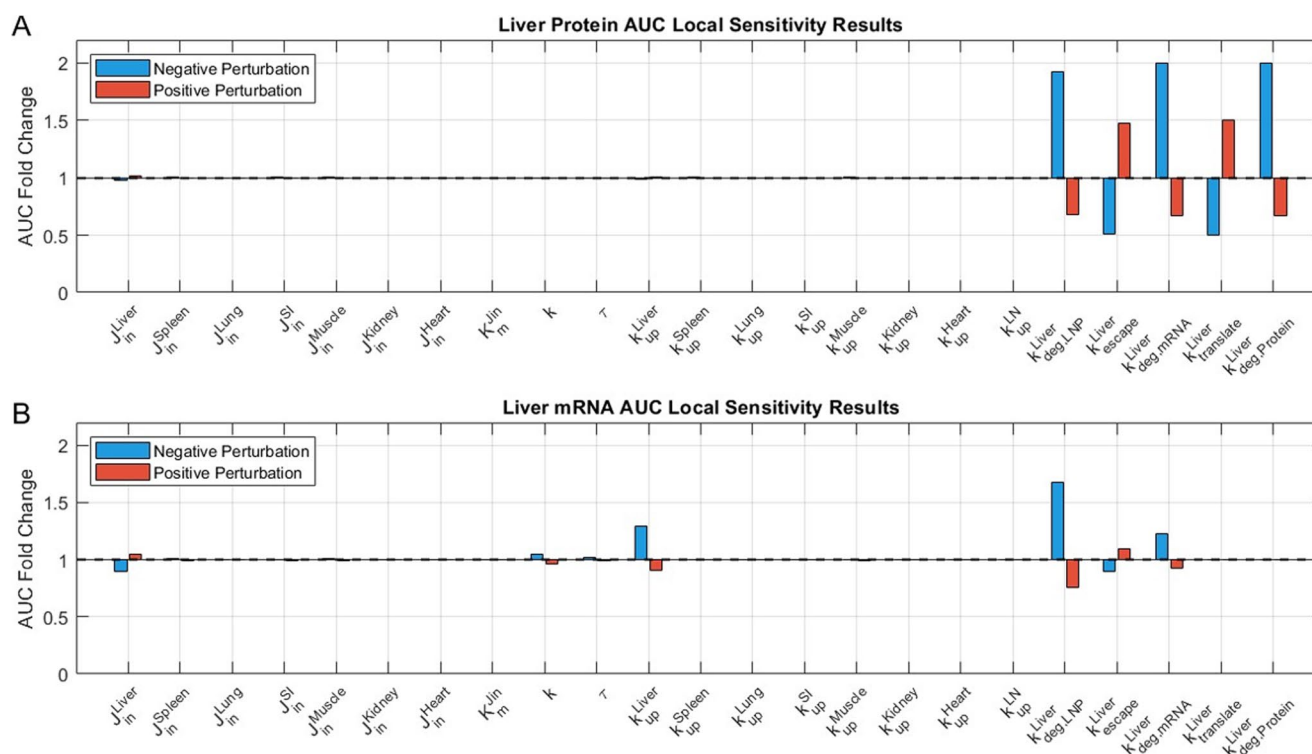
### Local and global sensitivity analysis of the model to identify important mechanistic parameters influencing mRNA and protein dynamics

A local sensitivity analysis was conducted to identify parameters impacting mRNA and protein dynamics in the liver. Each parameter was individually increased or decreased by 50%, and the resulting fold change in mRNA/protein AUC from the baseline simulation was plotted (Fig. 5). The results indicate that the most sensitive parameters influencing liver protein AUC are endosomal degradation rate ( $k_{deg,LNP}^{Liver}$ ), mRNA degradation rate ( $k_{deg,mRNA}^{Liver}$ ), and protein degradation rate ( $k_{deg,protein}^{Liver}$ ), with changes of up to 2-fold in protein AUC. Other notable parameters include the mRNA escape rate ( $k_{escape}^{Liver}$ ) and protein translation rate ( $k_{translate}^{Liver}$ ), which lead to ~1.5-fold change in the protein AUC. Liver protein AUC was not appreciably influenced by the perturbations in the remaining parameters.

On the other hand, liver mRNA AUC was relatively sensitive to LNP degradation rate (~1.7-fold change in mRNA AUC as a result of 50% reduction in LNP degradation rate) and to mRNA degradation and liver cellular uptake rate (~1.3-fold change in mRNA AUC following 50% change in these parameters).

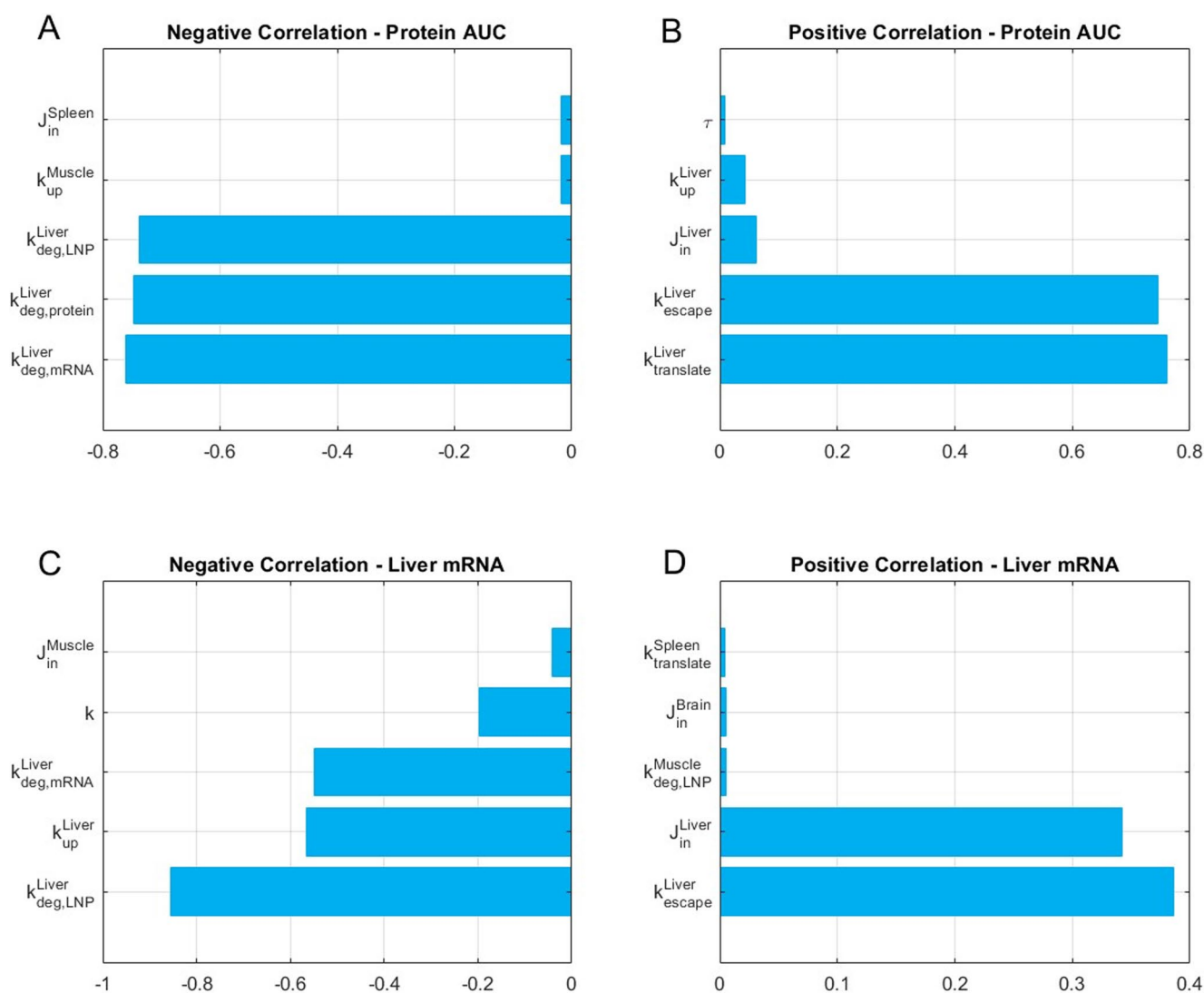
Positive and negatively correlated parameters influencing liver protein and mRNA AUC were ranked and plotted (Fig. 6). The results demonstrate that liver protein exposure was negatively correlated with mRNA degradation rate, protein degradation rate, and endosomal degradation rate (Fig. 6A), aligning with the local sensitivity analysis results. The most positively correlated parameters were the protein translation and mRNA escape rates (Fig. 6B).

On the other hand, liver mRNA exposure was negatively correlated with the LNP degradation rate, followed by the liver cellular uptake rate and parameters related to mRNA stability (Fig. 6C). Conversely, liver mRNA exposure was positively correlated with the mRNA escape rate and liver tissue influx rate (Fig. 6D). Interestingly, liver protein AUC was much less sensitive to the influx rate indicating that once the mRNA reach the liver tissue, the most significant parameters for protein exposure are LNP and mRNA stability rates and the translation rate.



**Fig. 5** Local sensitivity study of liver protein AUC (area under the curve) (A) and liver mRNA AUC (B). Each of the parameters was varied by 50% (decreased or increased), and corresponding fold changes in mRNA/protein AUC from the baseline were plotted. Red bars

indicate the fold change due to the increase in the parameter by 50%, and the blue bars indicate the fold change as a result of the parameter reduction by 50%. mRNA dose is fixed to 2 mg/kg and the infusion time is 10 min



**Fig. 6** Global sensitivity analysis using the PRCC method. (A) Ranking of the negatively correlated parameters for the liver protein profile, (B) Ranking of the positively correlated parameters for the liver protein profiles, (C) ranking of negatively correlated parameters for

the liver mRNA profiles, (D) positive correlation ranking for the liver mRNA profiles. mRNA dose and infusion time are fixed to 2 mg/kg and 10 min respectively

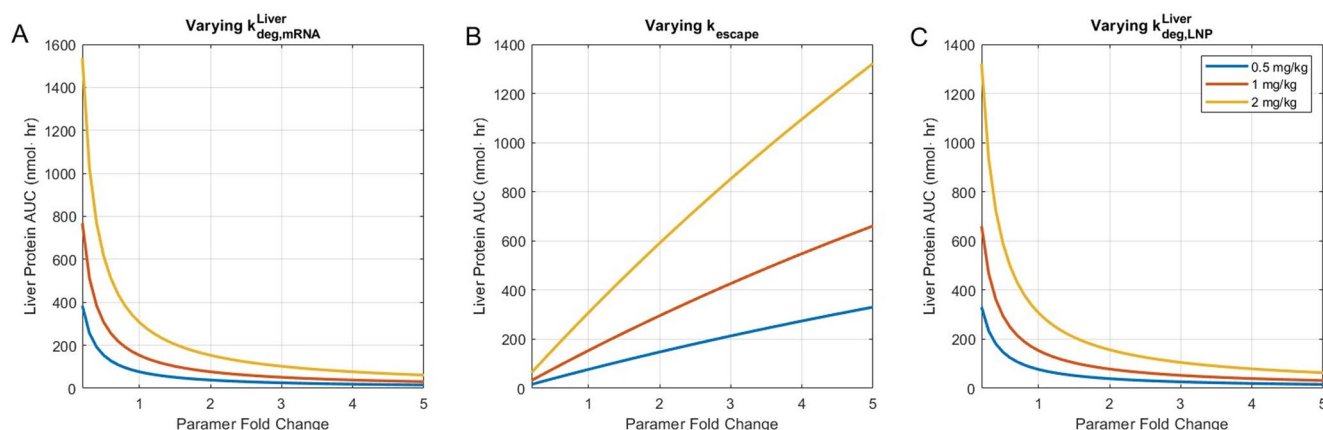
### Potential Key mRNA design parameter investigation

The model was then used to explore how key parameters related to mRNA-LNP treatment design may influence total protein production within the liver (Fig. 7). Three key design factors were tested: mRNA degradation rate, mRNA escape rate, and endosomal degradation rate. Each parameter was varied from 0.2- to 5-fold of the baseline value at three different dose levels (0.5, 1, and 2 mg/kg). Simulation results suggest that an increase in mRNA degradation rate substantially reduces protein production in the liver in a highly nonlinear manner (Fig. 7A). The impact on protein AUC appears to plateau above ~3 fold increase in the degradation rate.

The relationship between protein AUC and mRNA escape rate is linear at the three dose levels, indicating that

liver protein exposure may be tunable through changes in the mRNA escape rate (Fig. 7B). Protein AUC changes in a highly nonlinear in response to increase in LNP degradation rate. The decrease in protein AUC seems to saturate at high enough LNP degradation rates. These simulations appoint to a potential threshold like behavior in modulation of liver protein AUC by the mRNA and LNP degradation.

A Hill equation and a probability function for receptor availability were used in tissue influx transport to account for receptor-mediated transport through the endothelial layer, characterized by a saturation constant ( $K_m^{Jin}$ ) and receptor depletion rate ( $k_{receptor}^{Liver}$ ). Utilizing this equation allowed the model to better describe the rapid mRNA-LNP deposition across different tissue sites at the beginning of treatment. To further investigate how these parameters



**Fig. 7** Exploration of key model parameters. Liver protein AUC under three different doses (0.5, 1 and 2 mg/kg) was plotted by varying (A) liver mRNA degradation rate, (B) mRNA escape rate, and (C) liver endosomal degradation rate. Infusion time is fixed to 10 min

influence receptor-mediated tissue influx, the fold change in liver protein AUC from baseline was plotted as a function of  $k_{receptor}^{Liver}$  and influx rate  $J_{in}^{Liver}$  in a heatmap (Fig. 8A). Simulation results demonstrate that both receptor depletion and tissue influx rates modestly influence protein AUC in the liver. The more pronounced effect was on the  $t_{max}$  of protein PK in the liver. Increasing  $k_{receptor}^{Liver}$  from 0.2 to 5 fold of the base value shifts the  $t_{max}$  from ~13 h to ~28 h and reduces the  $C_{max}$  by ~40%. Conversely, increasing  $J_{in}^{Liver}$  from 0.2 to 5 fold of the base value shifts the  $t_{max}$  from ~28 h to ~13 h and increases  $C_{max}$  by approximately 1.8-fold.

### Projection of human clinical mRNA PK from rat

The calibrated PBPK model in rat was scaled to the human level to demonstrate inter-species translatability. Digitized clinical trial data from a study of an mRNA encoded monoclonal antibody against Chikungunya virus was used to evaluate the capability of the model for inter-species translation [5]. Virtual patients were generated to conduct a virtual clinical trial following the dosing regimens reported in the data: a single mRNA dose of 0.1 mg/kg, 0.3 mg/kg, or 0.6 mg/kg, as well as a two-dose regimen of 0.3 mg/kg, with the second dose administered on day 7. The infusion rate was fixed to 2 h consistent with the clinical protocol. Plasma PK time courses for the four dosing regimens were simulated using virtual patient data, and the resulting 95th percentile profiles were compared with the PK data (Fig. 9).

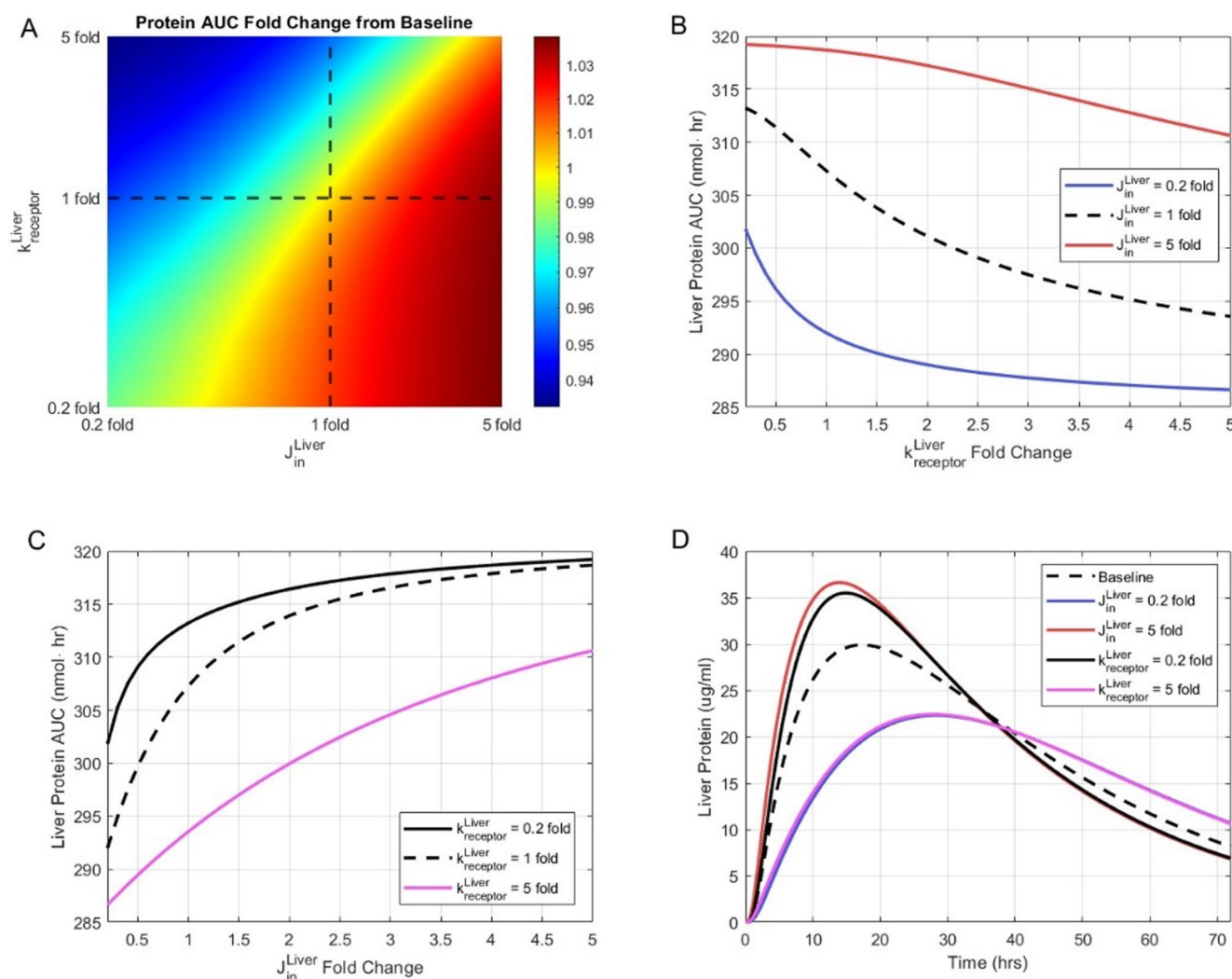
The human projection showed good agreement with the clinical data. The model adequately predicted the mRNA clearance profile, as shown by the overlap between the slope of the human data and the predicted percentile range from the virtual population simulation. Additionally, the model accurately predicted the  $C_{max}$  for the 0.6 mg/kg dose (Fig.

9C) and adequately captured the  $C_{max}$  for the lower dose levels (Figs. 9A, B and D). Additional sensitivity analyses were performed on the allometric scaling coefficients for tissue infiltration and cellular uptake (Figure S4). Perturbation of the tissue infiltration scaling exponent, as expected, moderately impacted the magnitude of systemic mRNA exposure. In contrast, perturbation of the cellular uptake scaling exponent had minimal impact on Cmax but showed some influence on the terminal clearance profile.

### Interplay of Kupffer cells and hepatocytes in shaping liver protein exposure

The human-scaled model, validated with clinical plasma PK data, was then used to explore the role of Kupffer cells in modulating protein expression within the liver tissue. Maximum protein expression in hepatocytes, Kupffer cells, and total liver tissue was compared by varying the cellular uptake rate into hepatocytes ( $k_{up}^{Hep}$ ) and Kupffer cells ( $k_{up}^{Kup}$ ), either together or separately (Fig. 10). When assuming the same cellular uptake rate for hepatocytes and Kupffer cells ( $k_{up}^{Hep} = k_{up}^{Kup} = k_{up}^{Liver}$ ), total liver protein  $C_{max}$  increases as the cellular uptake rate increases (Fig. 10A). However, when comparing  $C_{max}$  in hepatocytes vs. Kupffer cells, Kupffer cells exhibit a continuous near linear increase in protein  $C_{max}$ , while hepatocyte protein expression is saturated after initial increase as a function of the uptake rate (red solid line vs. blue solid line). The results suggest that LNP uptake into the Kupffer cells may be tuned by the changes in the uptake rate while the hepatocyte is relatively insensitive to variations in the uptake rate once a threshold rate of uptake is reached.

We next investigated the change in protein  $C_{max}$  in different cell types by individually varying the cellular uptake rates,  $k_{up}^{Hep}$  and  $k_{up}^{Kup}$  (Fig. 10B). For a given value of  $k_{up}^{Kup}$



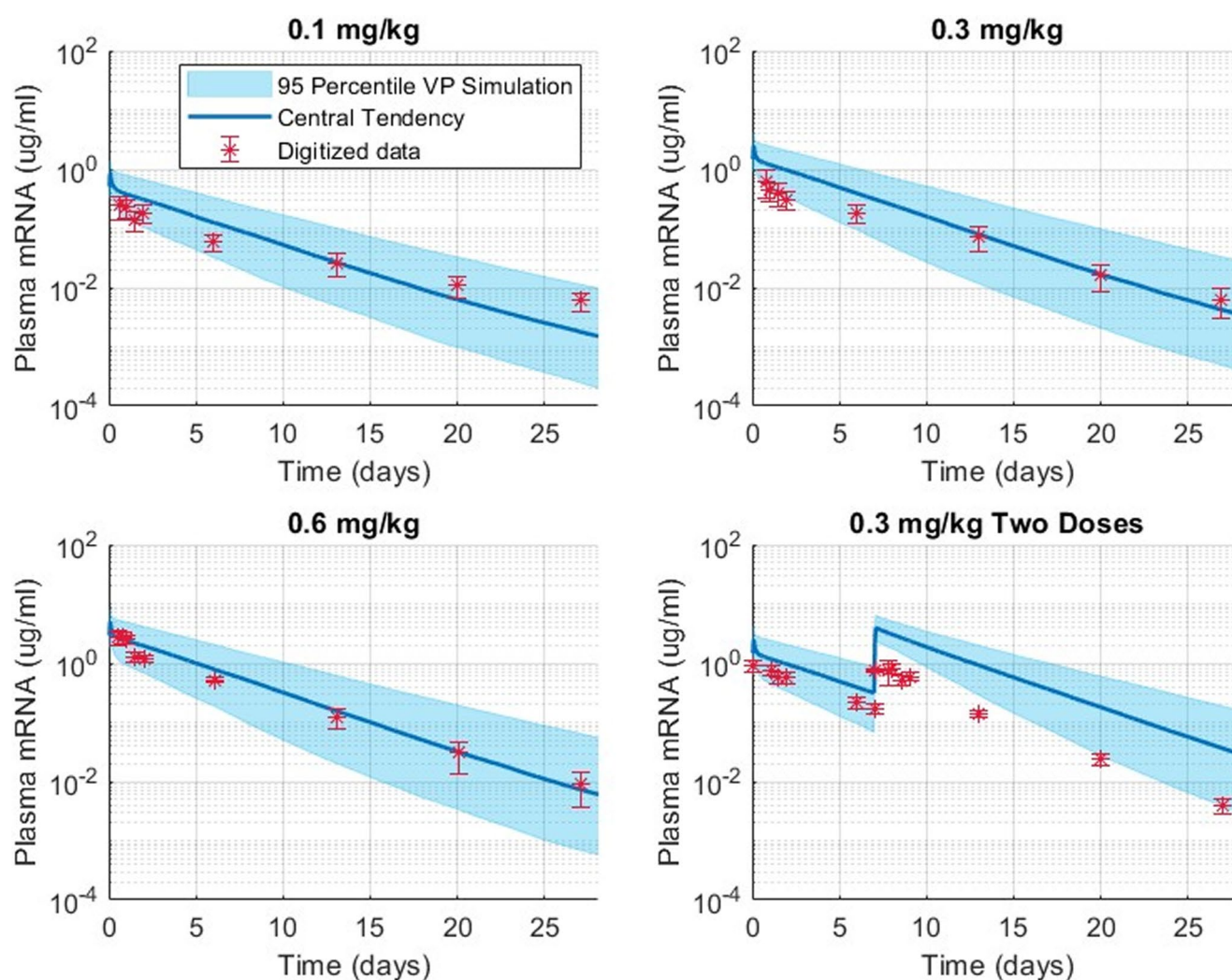
**Fig. 8** Exploring the impact of  $k_{receptor}^{Liver}$  (receptor depletion rate) during liver tissue influx and corresponding tissue influx rates on protein production. **(A)** Two parameter heatmap of liver protein AUC fold change from baseline. Liver influx rate on x axis in varied from 0.2 to 5 folds, and  $k_{receptor}^{Liver}$  value of liver influx on y axis is varied from 0.2 to 5 folds. Base value of  $k_{receptor}^{Liver}$  and  $J_{in}^{Liver}$  are indicated by horizontal and vertical dashed black lines respectively. **(B)** Liver protein AUC

under three different liver tissue influx rates.  $J_{in}^{Liver}$  was varied from 0.2 to 5 folds, **(C)** Liver protein AUC under three different  $k_{receptor}^{Liver}$  values. Receptor depletion rate was varied from 0.2 to 5 folds. **(D)** Liver protein time courses, varying liver influx rate and  $k_{receptor}^{Liver}$  by 0.2 and 5 folds respectively. Simulations for baseline parameter values are shown as dashed black line. mRNA dose and infusion time are fixed to 2 mg/kg and 10 min respectively

, increase in hepatocyte uptake rate resulted in increase in hepatocyte protein expression as expected. However, for a given value of hepatocyte uptake, the encoded protein  $C_{max}$  in hepatocytes tended to only modestly increase in response to reduction in the rate of uptake into Kupffer cells. Similarly, hepatocyte uptake did not appreciably perturb protein  $C_{max}$  in Kupffer cells, and that the encoded protein  $C_{max}$  in Kupffer cells was highly modulated by Kupffer cell LNP uptake rate (Figs. 10C). Total protein exposure in the liver was strongly modulated by both Kupffer cells and hepatocytes and both cell types were critical in modulation of liver tissue protein exposure in response to IV administration of mRNA-LNP (Fig. 10D).

## Discussion

This work presents a translational, whole-body PBPK platform for mRNA-LNP therapeutics that brings together three features relevant to establishing a translational workstream for mRNA therapeutics: (i) receptor-mediated, saturable trans-endothelial transport; (ii) intracellular trafficking with explicit endosomal and mRNA degradation and escape; and (iii) inter-species scaling from rat to human. Calibrated against a comprehensive rat biodistribution dataset containing both mRNA and translated protein and qualified with prediction of human plasma PK for an mRNA-encoded monoclonal antibody, the model closes a critical



**Fig. 9** Human mRNA PK projection. Scaled model was used to recreate the virtual clinical trial with (A) 0.1 mg/kg, (B) 0.3 mg/kg, (C) 0.6 mg/kg, and (D) 0.3 mg/kg with two doses. The predicted 5–95 percentile plasma mRNA PK was shown by the light blue shaded region

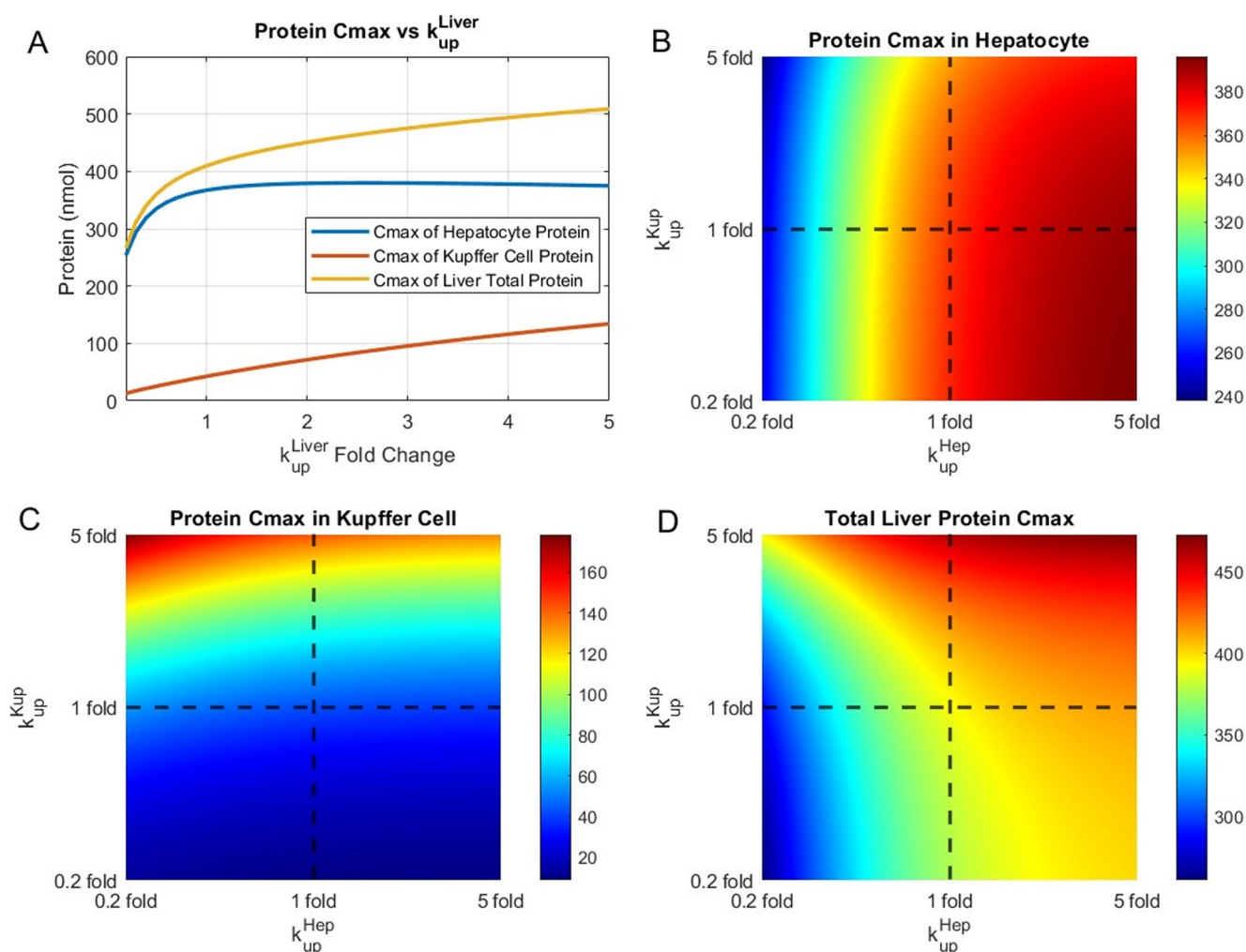
and predicted central tendency was shown by the solid blue line. The simulation results were compared against the published clinical trial data illustrated by the red asterisk

translational gap between empirical preclinical observations and clinical human projections. Beyond reproducing observed kinetics, the platform yields design-relevant insights into how LNP and mRNA properties shape tissue deposition and functional protein expression.

The early tissue and plasma kinetics necessitated a transport process faster than simple perfusion-limited exchange [22]. Introducing a saturable infiltration term with dynamic receptor availability captured the rapid deposition observed within minutes of dosing and reconciled the sharp early mRNA peaks across organs [27, 28]. While the present formulation is phenomenological, it is mechanistically consistent with a scenario in which circulating LNPs—conditioned by their protein corona—interact with endothelial and parenchymal receptors that can transiently saturate during high circulating concentrations [29]. In this sense, the

receptor-availability state variable serves as a surrogate for multiple microscopic processes (binding, internalization, recycling) that remain difficult to quantify *in vivo* but are essential to accurately capture the observed dynamics [30]. A rapid decline in plasma mRNA concentrations between 48 and 72 h was observed in the rat data but was not apparent in the available human data. Because late-phase mRNA kinetics seem to contribute minimally to overall protein production, a pragmatic modeling approach was adopted rather than introducing additional mechanistic complexity solely to improve the description of the rat data. This observation may warrant further investigation in future studies.

Local and global sensitivity analyses consistently suggest an important implication for therapeutic design: hepatic protein exposure is dominated by parameters determining intracellular dynamics of mRNA-LNP, specifically endosomal



**Fig. 10** Maximum protein expression ( $C_{max}$ ) at different cellular types in the liver with varying LNP cellular uptake rate. **(A)** Protein  $C_{max}$  in hepatocyte, Kupffer cell, and total liver while varying  $k_{up}^{Hep}$  and  $k_{up}^{Kup}$  simultaneously from 0.2 to 5 fold of the base value, **(B)** 2D color map of protein  $C_{max}$  within hepatocyte, varying  $k_{up}^{Hep}$  and

$k_{up}^{Kup}$  from 0.2 to 5 folds. **(C)** 2D color map of protein  $C_{max}$  within Kupffer cell, varying  $k_{up}^{Hep}$  and  $k_{up}^{Kup}$  from 0.2 to 5 folds. **(D)** 2D color map of total liver protein  $C_{max}$ , varying  $k_{up}^{Hep}$  and  $k_{up}^{Kup}$  from 0.2 to 5 folds

degradation, mRNA stability, and translation. Interestingly, liver exposure is less sensitive to increases in LNP tissue influx consistent with previously published modeling results [20]. Increasing mRNA escape from endosomes can linearly tune protein exposure over the explored range, whereas increasing endosomal or mRNA degradation produces strongly nonlinear response with threshold-like behavior. This result is consistent with available evidence demonstrating the critical role of mRNA stability and escape in modulating the encoded protein exposure, even suggesting mRNA escape as a critical determinant in designing optimal LNP based mRNA therapeutics [31, 32]. These findings are consistent with several well-known empirical levers that could be modulated - ionizable lipid pKa [33] and helper-lipid composition (endosomal escape) [34], UTR/codon design and chemical context (mRNA stability), and 5'

structural elements (translation) [35] - and the PBPK framework could provide a quantitative way to potentially guide the prioritization of which lever will matter most under specific dosing constraints. We note that the sensitivity analyses were performed at 2 mg/kg dose, where some process may begin to saturate. Thus, the relative sensitivities should be viewed as dose-dependent and primarily hypothesis generating. In addition, the model was specifically calibrated to the dataset reported by Ci et al., and further re-calibration and sensitivity analyses would be necessary to assess the relative importance of key mechanistic steps for LNPs with different formulations.

The liver microenvironment analysis using the PBPK model highlights potentially important role of cell-type competition in modulating liver protein expression [23]. Simulations show that increasing LNP uptake into Kupffer

cells produces a near-linear rise in Kupffer-cell protein but exerts a modest “sink” effect that can limit interstitial availability and influence hepatocyte expression beyond a threshold consistent with available experimental evidence highlighting the role of Kupffer cells in shaping LNP hepatic distribution [36, 37]. Conversely, once hepatocyte uptake increases beyond a threshold level, further gains in hepatocyte encoded protein  $C_{max}$  tend to saturate. This cell-type interplay may explain why formulations that reduce uptake in Kupffer cells (without compromising overall hepatic delivery) can yield improvements in hepatocyte protein and, by extension, in secreted-protein pharmacology [38]. The same framework can be applied to purposefully direct delivery toward other cellular targets (e.g., macrophage or endothelial cells) when that is the pharmacological intent.

The rat-to-human translation adopted standard allometry for tissue influx rate and cellular uptake rate (0.75 and  $-0.25$  exponents, respectively) and reproduced the human plasma mRNA clearance phase and peak timing across several doses. Small deviations in  $C_{max}$  at lower doses likely reflect population heterogeneity (body weight, infusion mixing, and inter-individual variability in cellular uptake/clearance) that was only partially represented in the virtual cohort. The model also predicts a higher  $C_{max}$  at the second dose under a repeated dosing regimen, which is attributed to reduced tissue infiltration due to receptor depletion following the initial dose. However, the model was calibrated using single-dose rat data. Incorporating additional preclinical data across multiple dose levels would improve the model’s ability to capture nonlinear PK behavior and enhance prediction accuracy. Nonetheless, the model is well qualified to be used for first-in-human (FIH) projections with the opportunity to refine the projections based on emerging clinical data.

From the standpoint of translational biomarkers, the PBPK platform makes it straightforward to link mechanistic assumptions to measurable surrogates. For example, if endosomal degradation is a dominant limiter for a given construct, the model predicts pronounced shifts in protein exposure with small changes in escape kinetics—suggesting that modest differences in ionizable lipid composition may manifest as outsized differences in circulating protein (for secreted payloads) or tissue PD (for intracellular targets). Similarly, when the model is used to generate “what-if” scenarios for receptor saturation, it predicts changes on very early plasma/tissue mRNA time-course, implying that sampling around infusion could be informative for deconvolving transport vs. intracellular rate limiting steps in FIH studies [20].

Several assumptions in this PBPK framework warrant future refinement. First, the  $K_m^{Jin}$  governing saturable infiltration was assumed uniform across tissues and species; in reality, receptor density, accessibility (fenestration), and corona composition are tissue- and state-dependent [39–41].

Second, a single (tissue-invariant) mRNA escape rate was used; endosomal pH dynamics and trafficking pathways likely vary by cell type and physiological context. Third, cellular uptake and translation efficiency was assumed equal between hepatocytes and Kupffer cells; differences in ribosomal load, innate immune activation, or RNA modification sensing could introduce cell-type-specific mRNA translation rates [42, 43]. While this assumption was made to maintain model conceptual simplicity given available data, additional experimental data would be needed to better characterize potential differences between these cell types and refine the model accordingly. Fourth, the lymphatic representation is simplified and does not yet incorporate size-dependent reflection or dynamic interstitial pressure - factors that could influence extrahepatic delivery. Finally, route-of-administration effects beyond IV (e.g., IM, SC) and disease states (fibrosis, inflammation) were not modeled and remain important extensions for clinical relevance in different populations.

These limitations point to concrete needs for more quantitative data in different systems. Quantitative receptor measurements (density, turnover) in relevant vascular beds, time-resolved endosomal pH/escape readouts, and single-cell resolved mRNA/protein trajectories would enable replacement of phenomenological terms with mechanistic modules. Complementary *in vitro* studies characterizing species-dependent cellular uptake or endosomal processing would further enhance model validation and mechanistic resolution. Imaging or tracer studies that capture the earliest time points could calibrate the saturation term more tightly. On the clinical side, paired early plasma sampling with minimally invasive tissue or soluble-protein biomarkers could help resolve transport vs. intracellular steps in individual patients. Together, such data would allow the platform to transition from a robust translational workhorse to a precision-modeling tool for patient- and indication-specific optimization. Although the current framework was calibrated using limited dataset, it was designed to be modular and data-driven, allowing systematic refinement as additional preclinical, clinical, or *in vitro* data becomes available. Importantly, application to LNPs with different chemistries or formulations would require appropriate recalibration and validation, which are beyond the scope of the present work.

Beyond liver centric applications, the same framework can be used to (i) explore extrahepatic targeting via alternative lipid compositions and targeting ligands; (ii) simulate complex modalities such as mRNA-encoded bi-specifics or cytokines with feedback on clearance or distribution; and (iii) integrate with QSP pharmacodynamic models to translate tissue protein time-courses into functional PD (e.g., enzyme replacement, immune activation). Because the model already operates at the organ and cellular-compartment levels, these extensions are largely modular: a

receptor-level transport block can be dropped into specific tissues, or a PD block (e.g., target engagement, signal transduction) can be integrated into protein output in a cell-type-resolved manner.

In sum, the platform provides a coherent quantitative scaffold for hypothesis testing, construct prioritization, and FIH planning. It helps identify which design knobs (lipid chemistry, RNA engineering, dose/infusion) could efficiently impact the clinical exposure and pharmacological response, and it highlights where additional biological characterization will likely offer the greatest return on experimental investment.

## Conclusion

We developed and qualified a translational whole-body PBPK model for mRNA-LNP therapeutics that: (i) mechanistically captures saturable endothelial transport and intracellular processing; (ii) reproduces organ-resolved mRNA and protein kinetics in rat; and (iii) scales to human to recapitulate plasma mRNA PK across clinically relevant doses. Sensitivity analyses identify endosomal degradation, mRNA stability, and translation as the principal determinants of hepatic protein exposure once initial delivery is achieved, and cell-type analyses quantify how potential Kupffer-cell “sink” effects may constrain hepatocyte expression. Practically, these results prioritize escape/stability engineering over further increases in gross hepatic deposition, and they provide a rational basis for potentially tuning LNP properties to favor desired liver cell types.

Immediate applications include FIH dose-range selection, infusion-time optimization, and biomarker strategy (sampling windows and PD surrogates). Looking forward, embedding receptor-level transport, route-specific input functions (IM/SC), and disease-state physiology, together with richer cellular data, will further extend applicability beyond liver-focused indications. By marrying mechanistic modeling with translational utility, this platform offers a generalizable foundation for designing, de-risking, and advancing next-generation mRNA medicines.

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**Author contributions** K.M. and H.B. developed the PBPK model. K.M. performed all simulations and generated all figures. All authors contributed to the study design, wrote the manuscript, and reviewed and approved the final manuscript.

**Data availability** The model equations and parameters are provided in the supplementary Material associated with this article. The experimental data used for model calibration were obtained from published study, which is cited in the manuscript.

## Declarations

**Ethics** Not applicable.

**Consent to participate** Not applicable.

**Consent to publish** Not applicable.

**Competing interests** The authors declare no competing interests.

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